Definitions

Closed Sets: A set $S \subseteq \mathbb{R}^n$ is closed if for every convergent sequence $\{x_k\} \subseteq S$, its limit lies in S; that is, if $x_k \to z$ then $z \in S$.

Open Sets: A set $S \subseteq \mathbb{R}^n$ is open if for every point $x \in S$, there exists a radius $r > 0$ such that the open ball $B_r(x)$ is contained in S; that is, $B_r(x) \subseteq S$.

Bounded Sets: A set $S \subseteq \mathbb{R}^n$ is bounded if there exists $M > 0$ such that $\|x\| \leq M$ for all $x \in S$.

Compact Sets: A set $S \subseteq \mathbb{R}^n$ is compact if it is both closed and bounded (Heine–Borel Theorem).